Infinite NPA depth at minimal semidefinite complexity

Anubhav Chaturvedi[1]

[1] *Division of Quantum Optics and Information, Institute of Theoretical Physics and Astrophysics, Faculty of Mathematics, Physics and Informatics, University of Gdańsk, 80-308 Gdańsk, Poland*

(Dated: 27 October 2026)

The quantum region of the CHSH value and the sum of two marginal averages is not recovered by any finite level of the Navascués–Pironio–Acín hierarchy. We give an exact finite semidefinite lift of this same region. The proof reduces every supporting inequality to equal measurement overlaps. At fixed product of the overlap parameters, the determinant is affine in their sum. In the nonclassical range the coefficient is nonnegative, so equal overlaps minimize it; the remaining bounds are local. The resulting univariate quartic matrix depends affinely on the supporting inequality and gives an explicit lift with blocks of orders five and four. The region has semidefinite extension degree two, the smallest possible for a nonpolyhedral set. Thus the family that requires unbounded NPA depth has a finite representation of minimal semidefinite block order.

Alice and Bob each choose between two measurements with outcomes ±1. Their marginal averages are $a_x = \langle A_x \rangle$ and $b_y = \langle B_y \rangle$, and their correlators are $E_{xy} = \langle A_x B_y \rangle$. Retain two observed numbers:

$$M = a_0 + b_0, \qquad S = E_{00} + E_{01} + E_{10} - E_{11}. \quad (1)$$

The second is the CHSH value [1]. Let $\mathcal{W}$ be the set of pairs (M, S) obtainable in quantum theory. All quantum correlations are allowed, including those with unequal marginal averages.

In our preceding Letter [2], we showed that no fixed level of the Navascués–Pironio–Acín (NPA) hierarchy [3, 4] recovers this region. For every finite level, symmetric doubly tilted inequalities $S + \alpha M \leq \omega_{\mathrm{Q}}(\alpha)$ are overestimated sufficiently close to the local threshold $\alpha = 1$. We now construct one exact finite semidefinite representation of the entire region $\mathcal{W}$.

Theorem 1. *The compact convex quantum region $\mathcal{W}$ has an affine semidefinite lift with blocks of orders five and four and sixteen auxiliary variables. If $\mathcal{W}_L$ is its level-L NPA outer approximation, then*

$$\mathcal{W} \subsetneq \mathcal{W}_L \quad \textit{for every finite } L, \qquad \mathrm{sxdeg}(\mathcal{W}) = 2. \quad (2)$$

A semidefinite lift describes a set by affine constraints and positive semidefinite (PSD) matrices in its observed and auxiliary coordinates. The extension degree sxdeg is the smallest possible largest block order among such finite lifts. We construct the explicit $5, 4$ lift below; the optimal order two follows from a planar representation theorem.

Two measurement angles, one determinant.—A compact convex region is the intersection of its supporting half-planes. It therefore suffices to decide, for every real h, α, λ, whether

$$hS + \alpha M \leq \lambda \qquad \text{throughout } \mathcal{W}. \quad (3)$$

For fixed measurements this is equivalent to positivity of $\lambda I - H$, where $H = h\,\mathrm{CHSH} + \alpha(A_0 + B_0)$: its expectation in any state is the slack in Eq. (3). Jordan's lemma reduces the measurements to two-qubit blocks. Choosing local axes gives $A_0 = B_0 = Z$ and

$$A_1 = cZ + \sqrt{1-c^2}X, \quad B_1 = dZ + \sqrt{1-d^2}X. \quad (4)$$

Here $c, d \in [-1, 1]$ are the cosines of the two local measurement angles. Deterministic strategies attain $\ell = 2|h| + 2|\alpha|$, so $\lambda \geq \ell$ is necessary.

Above this local bound, the first three elementary symmetric functions of the eigenvalues of $\lambda I - H$ are nonnegative. Thus its determinant decides positivity [5]. Indeed, if the determinant is also nonnegative, $\det(rI + \lambda I - H)$ has nonnegative coefficients and no positive root; a negative eigenvalue would give such a root.

Put $x = (1-c)/2$ and $y = (1-d)/2$. The determinant has the form

$$D(x, y) = F(xy) + 64h^2(4h^2 - \alpha^2)xy(x + y), \quad (5)$$

where F is a polynomial in the product alone. For $|\alpha| \leq |h|$, the coefficient of $x+y$ is nonnegative. At fixed product, $x + y \geq 2\sqrt{xy}$, with equality at $x = y = \sqrt{xy}$. Hence

$$D(x, y) \geq D(\sqrt{xy}, \sqrt{xy}).$$

Positivity at equal overlaps therefore implies positivity at every pair of overlaps. This elementary minimization removes one measurement angle.

The other directions, $|\alpha| \geq |h|$, have quantum support ℓ. The one-sided bounds $\pm\mathrm{CHSH} + 2\tau A_0 \leq 4I$ and $\pm\mathrm{CHSH} + 2\tau B_0 \leq 4I$, $\tau = \pm1$, follow from the critical tilted-CHSH sum of squares [5]. Average the two with the required signs, multiply by $|h|$ and add the marginal slack of weight $|\alpha| - |h|$. This proves $H \leq \ell I$, attained by equal-overlap deterministic strategies. Equal overlaps thus decide *all* supporting lines.

One angle, one finite lift.—At $c = d$, the singlet decouples with eigenvalue $-2hc$. On the remaining three Bell states, set $c = (1-t^2)/(1+t^2)$ and multiply the slack by $(1+t^2)^2$. With

$$p_0 = (1+t^2)^2, \qquad p_1 = 1 - t^4, \qquad p_2 = 1 + 4t^2 - t^4,$$

the resulting quartic matrix is

$$K(t) = \begin{pmatrix} \lambda p_0 - 2hp_1 & -2\alpha p_0 & 0 \\ -2\alpha p_0 & \lambda p_0 - 2hp_2 & -8ht^3 \\ 0 & -8ht^3 & \lambda p_0 + 2hp_2 \end{pmatrix}. \quad (6)$$

Its value at zero and its leading coefficient force $\lambda \geq \ell$, which also controls the singlet. The endpoint $c = -1$ follows by continuity, and the parity $K(-t) = RK(t)R$, $R = \mathrm{diag}(1, 1, -1)$, allows all real t. Consequently

$$\max_{\mathcal{W}}(hS + \alpha M) \leq \lambda \quad \Longleftrightarrow \quad K(t) \succeq 0 \quad \forall t \in \mathbb{R}. \quad (7)$$

A PSD real univariate matrix polynomial is a polynomial matrix sum of squares, including when singular [6]. For a quartic, every factor has degree at most two, since leading real squares cannot cancel. For $\xi \in \mathbb{R}^3$, the parity of K splits the quadratic features into

$$\begin{aligned} Z_+ &= (\xi_1, \xi_2, t\xi_3, t^2\xi_1, t^2\xi_2)^T, \\ Z_- &= (\xi_3, t\xi_1, t\xi_2, t^2\xi_3)^T. \end{aligned} \quad (8)$$

Thus Eq. (7) is equivalent to the polynomial identity

$$\xi^T K(t)\xi = Z_+^T G Z_+ + Z_-^T J Z_-, \qquad G \succeq 0, \quad J \succeq 0. \quad (9)$$

The blocks have orders five and four. Matching coefficients eliminates t and gives sixteen affine equations.

Since K is affine in *all* support coefficients, these equations describe the whole cone of valid inequalities. Semidefinite duality therefore gives a single membership lift:

$$\mathcal{W} = \left\{ (M, S) : \begin{array}{l} \exists z \in \mathbb{R}^{16},\ U(z), V(z) \succeq 0, \\ \ell(z) = 1,\ m(z) = M,\ s(z) = S \end{array} \right\}. \quad (10)$$

Here U, V are the adjoints of the coefficient map, and ℓ, m, s are the corresponding linear coordinate maps; all their rational entries are given in the Supplemental Material [5].

The lift also attains the boundary. The Gram pair

$$G_0 = \mathrm{diag}(1, 1, 2, 1, 1), \qquad J_0 = \mathrm{diag}(1, 2, 2, 1)$$

is positive definite and represents $(\lambda, \alpha, h) = (1, 0, 0)$. It supplies strict feasibility. Moreover, the normalization is $\mathrm{tr}(U(z)G_0) + \mathrm{tr}(V(z)J_0) = 1$, which bounds both PSD matrices and every auxiliary variable. The normalized dual image is compact, and its supporting half-planes are exactly Eq. (7). This proves equality in Eq. (10), including boundary points.

What the NPA obstruction measures.—Equal overlaps determine the support values, whereas a finite NPA certificate must represent the bound for independently chosen measurement angles. The preceding proof of nontermination uses precisely that two-variable freedom: near the local threshold, an NPA error bounded by a constant times the quantum violation above the local bound would force a shifted Motzkin polynomial to be a sum of polynomial squares [2]. The univariate squares in Eq. (9) arise *after* the support minimization and need not be identities in a fixed list of observable words. The full earlier argument is included in the companion proofs.

The optimal block order follows from the same boundary. For $f(\alpha) = \max_{\mathcal{W}}(S + \alpha M)$, the exact quantum bound [7] has the expansion [2]

$$f(1 - T) = 4 - 2T + \tfrac{4}{3}T^3 + O(T^4), \qquad T \downarrow 0. \quad (11)$$

A polygon would have affine support for α sufficiently close to one, contrary to Eq. (11). Thus $\mathcal{W}$ is nonpolyhedral. It is also closed, convex and semialgebraic by Eq. (10); Scheiderer's planar theorem [8] gives $\mathrm{sxdeg}(\mathcal{W}) \leq 2$. Order-one lifts are polyhedral, so $\mathrm{sxdeg}(\mathcal{W}) = 2$. The optimal-order representation is existential; the blocks $5, 4$ above are explicit.

Finally, physical relabellings give the same quantum image $\mathcal{W}$ for

$$\pi_{\chi,xy}(p) = \left(a_x + \chi_{xy} b_y,\ \sum_{ij} \chi_{ij} E_{ij} \right). \quad (12)$$

Here $\prod_{ij} \chi_{ij} = -1$. The eight CHSH sign patterns and four selected input pairs give 32 distinct planes. Intersecting any NPA relaxation with $\pi^{-1}_{\chi,xy}(\mathcal{W})$ for all of them retains every quantum correlation and gives exact support on all 32 planes. In particular, it resolves the entire family responsible for finite-level nontermination.

[1] J. F. Clauser, M. A. Horne, A. Shimony, and R. A. Holt, Proposed experiment to test local hidden-variable theories, Phys. Rev. Lett. **23**, 880 (1969).

[2] A. Chaturvedi, *No Finite NPA Level Characterizes the Complete Quantum Set in the Simplest Bell Scenario*, manuscript (2026).

[3] M. Navascués, S. Pironio, and A. Acín, Bounding the set of quantum correlations, Phys. Rev. Lett. **98**, 010401 (2007).

[4] M. Navascués, S. Pironio, and A. Acín, A convergent hierarchy of semidefinite programs characterizing the set of quantum correlations, New J. Phys. **10**, 073013 (2008).

[5] See the accompanying Supplemental Material for the complete finite-lift proof, explicit membership matrices, all signed asymmetric support branches, physical relabellings and the full earlier NPA proof.

[6] C. Hanselka and R. Sinn, Positive semidefinite univariate matrix polynomials, Math. Z. **292**, 83–101 (2019), Corollary 2.8.

[7] N. Gigena, E. Panwar, G. Scala, M. Araújo, M. Farkas, and A. Chaturvedi, Self-testing tilted strategies for maximal loophole-free nonlocality, npj Quantum Inf. **11**, 82 (2025).

[8] C. Scheiderer, Second-order cone representation for convex sets in the plane, SIAM J. Appl. Algebra Geom. **5**, 114–139 (2021), Theorem 0.1.